

Functional Requirements Specification (FRS)

1. DOCUMENT CONTROL

Field	Value
Application Name	Subsidy Delivery Partner (Pod Delivery)
Document Name	Functional Requirements Specification
Document ID	FRS-POD-001
Version	1.2
Status	Approved & Production Ready
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Date	2026-08-31

2. PURPOSE

The purpose of the Subsidy Delivery Partner application is to enable delivery personnel to securely and verifiably deliver subsidized agricultural goods to registered farmers. It solves the business problem of delivery fraud by enforcing strict verification mechanisms. The primary intended users are delivery partners. This FRS documents what the application must do, serving as the source of truth for Developers, QA Engineers, Product Owners, and Stakeholders.

3. SCOPE

3.1 In Scope

- Mobile OTP-based authentication and partner registration.
- Strict 8-step partner profile completion (Name filter, Email regex, PIN code auto-fetch, City/State, Address, Vehicle Type, Indian Vehicle Number format, 12-digit Aadhaar, Photo upload to Cloudinary).
- Viewing assigned pending and completed deliveries, plus on-the-fly farmer beneficiary creation.
- GPS-based geofenced delivery validation (600m radius constraint).
- Customer OTP verification at the time of delivery.
- On-device facial recognition (Google ML Kit / TFLite) matching the recipient to the registered farmer photo.
- Capturing multi-photo item proofs (1 to 5 photos) and optional video proof.
- Generating delivery invoices (PDF), auto-upload to Cloudinary, and native in-app viewing via Syncfusion PDF Viewer.

3.2 Out of Scope

- Admin web portal for assigning deliveries (handled centrally).
- Direct communication/chat with the customer.
- Payment gateway integration (focus is strictly on physical subsidy goods verification).
- App-based turn-by-turn route navigation (delegated via OS geo: map intents).

4. PRODUCT OVERVIEW

- **Application Purpose:** Secure verification and logging of subsidy deliveries.
- **Target Users & Roles:** Delivery Partners (Field Operatives).
- **Platforms:** Android (Flutter SDK client).
- **Major Modules:** Authentication, Profile (8-Step), Dashboard, Order Completion, In-App PDF Invoice Viewer.
- **External Systems & Cloud:** Railway Production Server (<https://pod-app-production-818a.up.railway.app>), MongoDB Atlas (pody_db), Cloudinary Media Storage, Indian Postal Pincode API, Google ML Kit.

High-Level Workflow:

Application Launch → Mobile Authentication (OTP: 1234) → Profile Completion (Mandatory 8 Steps) → Dashboard (Customer List) → Select Pending Customer → Order Completion (5 Gates: 600m GPS + Face Verify + OTP + Photos + Video) → Backend Cloudinary & MongoDB Sync → Success (Native PDF Receipt Inspection)

5. USER ROLES AND ACCESS

Role ID	Role	Description	Functional Access
UR-01	Delivery Partner	App user delivering goods.	Full app access (Auth, Profile, Deliveries, POD).
UR-02	Customer/Farmer	Recipient of goods.	N/A (Interacts in-person; does not use app).

6. APPLICATION NAVIGATION

Splash Screen
↓
Login/Register Screen (10-digit Phone + OTP)
↓
Profile Completion Screen (Mandatory 8-Step Form with Auto PIN Lookup)
↓
Dashboard (Pending / Delivered Tabs)
 ■■■■ Add Farmer Dialog (On-the-fly registration)
 ■■■■ Customer Details Screen
 ■ ■■■■ Order Completion Screen (5 Verification Gates)
 ■ ■■■■ Native PDF Viewer Screen (Syncfusion PDF Viewer)
 ■■■■ Logout Action

7. FUNCTIONAL MODULES

- 1. **AUTH** - Authentication & Registration
- 2. **PROF** - Delivery Partner Profile Management (Strict 8-Step Sequence)
- 3. **DASH** - Dashboard, Customer List & On-the-fly Beneficiary Creation
- 4. **ORD** - Order Completion, 5 Verification Gates & Native Invoicing

8. FUNCTIONAL REQUIREMENTS

Master Requirement Table

ID	Module	Requirement	Type	Pri	Source	Status
FR-AUTH-001	AUTH	Authenticate users via 10-digit phone & OTP.	Functional	P0	Code	Confirmed
FR-PROF-001	PROF	Mandate strict 8-step profile completion.	Business Rule	P0	Code	Confirmed
FR-PROF-002	PROF	Block numbers/special chars in Name field.	Validation	P0	Code	Confirmed
FR-PROF-003	PROF	Auto-fetch City/State on 6-digit PIN entry.	Integration	P0	Code	Confirmed
FR-PROF-004	PROF	Validate Indian vehicle number format.	Validation	P0	Code	Confirmed
FR-ORD-001	ORD	Enforce 600m geofence for delivery.	Validation	P0	Code	Confirmed
FR-ORD-002	ORD	Verify recipient face using ML Kit / TFLite.	Authorization	P0	Code	Confirmed
FR-ORD-003	ORD	Require customer 4-digit handover OTP.	Authorization	P0	Code	Confirmed
FR-ORD-004	ORD	Multi-photo (1-5) and video proof capture.	Functional	P0	Code	Confirmed
FR-ORD-005	ORD	Generate and view invoice via Syncfusion.	Functional	P0	Code	Confirmed

Detailed Requirements

Requirement ID: FR-ORD-001

- **Requirement Name:** Geofenced Delivery Validation (600m Constraint)
- **Requirement Type:** Validation | **Priority:** P0 | **Source:** Source Code Confirmed
- **Requirement:** The system shall prevent the submission of an order completion form if the device's current GPS location is more than 600 meters from the customer's registered coordinates.
- **Preconditions:** The user is on the Order Completion screen. Location permission is granted.
- **Input & Processing:** Device Lat/Lng and Customer Lat/Lng calculated via `Geolocator.distanceBetween()`.
- **Output & Behavior:** Displays green proximity banner if $\leq 600\text{m}$; displays warning and blocks submit if $> 600\text{m}$.

Requirement ID: FR-ORD-002

- **Requirement Name:** Facial Recognition Verification
- **Requirement Type:** Authorization | **Priority:** P0 | **Source:** Source Code Confirmed
- **Requirement:** The system shall verify the identity of the recipient by comparing a live captured photo against the registered farmer photo using on-device Google ML Kit / TFLite.
- **Input & Processing:** Live photo file, Downloaded reference photo. Extracts landmark embeddings and calculates cosine similarity.
- **Output & Behavior:** Match success enables submission. Mismatch displays 'Face mismatch!' error dialog.

Requirement ID: FR-PROF-001 to FR-PROF-004

- **Requirement Name:** Mandatory 8-Step Profile Onboarding & Real-Time Postal Lookup
- **Requirement Type:** Validation & Integration | **Priority:** P0 | **Source:** Source Code Confirmed
- **Sequence & Validation Rules:**
 1. **Full Name:** Restricted strictly to letters and spaces via `FilteringTextInputFormatter.allow(RegExp(r'[a-zA-Z\s]'))`. Digits/symbols are blocked.
 2. **Email:** Validated against RFC regex `^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`; spaces blocked.
 3. **PIN Code:** 6 numeric digits. Auto-triggers background lookup to `https://api.postalpincode.in/pincode/{PIN}`.
 4. **City & State:** Auto-filled from resolved postal data (read-only with manual unlock toggle).
 5. **Address:** Minimum 5 characters.
 6. **Vehicle Type:** Dropdown (Two Wheeler, Three Wheeler, Four Wheeler, Other).
 7. **Vehicle Number:** Auto-capitalized; validated against Indian vehicle registration format (e.g. DL 01 AB 1234 or BH Series).
 8. **Aadhaar Number:** Exactly 12 digits; UIDAI format (cannot start with 0 or 1: `^[2-9][0-9]{11}$`).
- **Profile Photo:** Captured and uploaded via multipart to Cloudinary.

9. UI FUNCTIONAL BEHAVIOR

- **OTP Input:** Automatically advances focus to the next field upon digit entry and retreats on backspace.
- **Form Submission:** Validates all mandatory fields, formatters, and regex rules before sending network requests.
- **Loading States:** Disables action buttons and displays a CircularProgressIndicator during API operations.
- **Location Banner:** Dynamically renders green (success, <=600m) or amber/red (warning, >600m) status on Order Completion.
- **PIN Auto-Lookup UI:** Shows an in-line progress spinner in the PIN field while resolving postal location.

10. WORKFLOWS

Order Completion Workflow:

Trigger: User taps 'Complete Delivery' → Precondition: Location & Camera permissions granted → System Validation: Real-time Haversine distance check ($\leq 600\text{m}$) → User Action: Enter 4-digit Customer Handover OTP → User Action: Capture Recipient Face (ML Kit Match) → User Action: Capture 1-5 Item Photos + Optional Video → Submit: Multi-part cloud upload to Cloudinary & Railway API → Success: Update MongoDB pody_db status to 'delivered' & Open Native In-App PDF Receipt.

11. INPUT AND OUTPUT REQUIREMENTS

Req ID	Input	Validation	Processing	Output
FR-AUTH-001	Mobile Number	Exactly 10 digits	Query MongoDB pody_db	Trigger SMS OTP
FR-PROF-001	Full Name	Letters & spaces only	Input filter block	Sanitized Name
FR-PROF-003	PIN Code	Exactly 6 digits	Indian Postal API	Auto City & State
FR-PROF-004	Vehicle Number	Indian registration regex	Auto-capitalization	Validated Vehicle
FR-PROF-001	Aadhaar Number	12 digits, no 0/1 prefix	UIDAI format check	Saved to Profile
FR-ORD-003	Customer OTP	Exactly 4 digits	Match record	Allow/Block Submit

12. VALIDATION REQUIREMENTS

ID	Field	Rule	Expected System Behavior
VR-001	Mobile Number	Length == 10, Digits only	Show error: 'Please enter a valid 10-digit mobile number.'
VR-002	Full Name	Alphabetical & spaces only, 2-50 chars	Keyboard blocks invalid keys; shows error if < 2 chars.
VR-003	Email Address	RFC regex format; no spaces	Show error: 'Enter a valid email address (e.g. name@example.com)'.
VR-004	PIN Code	Length == 6, Digits only	Show error: 'PIN code must be exactly 6 digits'.
VR-005	Vehicle Number	Standard Indian format / BH Series	Show error: 'Enter valid vehicle number (e.g. DL 01 AB 1234)'.
VR-006	Aadhaar Number	Length == 12, starts with 2-9	Show error: 'Enter valid 12-digit Aadhaar (cannot start with 0 or 1)'.
VR-007	Profile Photo	Mandatory (Not null)	Prevent submission; show 'Please upload a profile photo'.
VR-008	Item Photos	Count >= 1 and <= 5	Prevent submission; show 'Please capture at least 1 item photo'.

13. AUTHENTICATION & AUTHORIZATION

- **Login:** Authenticates via 10-digit mobile number.
- **OTP:** 4-digit numeric code required to verify login/registration (Default Testing OTP: 1234).
- **Registration:** New users provide Name, Mobile, and Aadhaar before OTP verification.
- **Session Handling:** Managed via JWT Bearer Access Token stored securely in flutter_secure_storage.
- **Unauthorized Access:** HTTP 401 responses clear stored tokens and redirect operative to the Login screen.

14. DATA FUNCTIONAL REQUIREMENTS

ID	Entity	Operation	Requirement Description
DR-001	DeliveryPartner	Create	Creates a new partner document in MongoDB pody_db upon OTP registration.
DR-002	DeliveryPartner	Update	Updates profile fields and Cloudinary profile photo URL upon Profile Completion.
DR-003	Farmer / Customer	Read	Retrieves beneficiaries assigned to delivery_partner_id (All/Pending/Delivered).
DR-004	Farmer / Customer	Create	Adds new beneficiary record on-the-fly with Cloudinary photo and GPS geotagging.
DR-005	DeliveryLog / Order	Update	Records delivery status, proofs (Cloudinary URLs), and invoice PDF reference.

15. ERROR AND EXCEPTION HANDLING

- **Network Unavailable:** Catches SocketException / TimeoutException; shows 'Unable to connect to server. Please check your internet.'
- **Location Permission Denied:** Disables proximity checking; shows 'Unable to fetch location. Please enable GPS and allow location permissions.'
- **Geofence Violation:** Prohibits delivery submission if > 600m; shows 'Action Denied: You are Xm away from the customer's registered location.'
- **Face Mismatch:** Blocks delivery completion if cosine similarity fails; shows 'Face mismatch! The person does not match the registered farmer.'

16. DEVICE PERMISSIONS

Permission	Functional Purpose	Request Trigger	If Granted	If Denied
Location	600m geofencing check.	App launch / Order Screen	Calculates distance.	Blocks order completion.
Camera	Face & item proofs, video.	Profile / Order Screen	Opens camera feed.	Blocks proof capture.

17. NETWORK BEHAVIOR

- **Online:** Connects to Railway production server (<https://pod-app-production-818a.up.railway.app>), fetches deliveries, resolves postal PIN codes, and streams multipart media to Cloudinary.
- **Resilience:** Network timeouts extended to 120 seconds for media uploads to accommodate low-bandwidth rural cellular coverage.

18. INTEGRATIONS

Integration	Purpose	Trigger	Input	Expected Output	Failure Behavior
Railway Backend	Data sync, Auth, POD	App actions	JSON / Multipart	JSON responses	Show error snackbar
Cloudinary	Photos, Videos, PDFs	Profile / Order submit	Media bytes	HTTPS CDN URLs	Show upload error
Indian Postal API	PIN code resolution	6th digit in PIN field	6-digit PIN	City & State strings	Allow manual entry
Google ML Kit	Face verification	Face capture step	Image files	Boolean match	Block submission
Geolocator	GPS distance check	Order Screen open	GPS sensor	Distance (meters)	Block submission

19. BUSINESS RULES

- **BR-001:** A user shall not be allowed to submit delivery unless all mandatory proofs (Face Photo Match, Item Photo, valid Customer OTP) are provided.
- **BR-002:** A delivery is strictly prohibited if the partner is > 600m away from the customer's registered coordinates.
- **BR-003:** A delivery is strictly prohibited if the captured recipient face does not match the registered reference photo.
- **BR-004:** A delivery partner cannot access orders until their 8-step profile is completely filled, verified, and saved.

20. SECURITY FUNCTIONAL REQUIREMENTS

- **Authentication:** Access strictly requires a valid mobile OTP and JWT bearer token signed with HS256.
- **Sensitive Data Masking:** Tokens and sensitive credentials are never displayed in clear text on the UI.
- **Storage:** Session tokens are persisted using OS-level encrypted storage (flutter_secure_storage).
- **Biometric Privacy:** Facial feature extraction runs on-device; unencrypted facial vectors are not transmitted.

21. NOTIFICATIONS

- **Status:** In-app reactive status alerts via custom Snackbars and Dialogs. Push notification integration is planned for future releases.

22. NON-FUNCTIONAL REQUIREMENTS

- **NFR-001 (Performance):** On-device facial recognition executes within 3 seconds on standard Android mobile hardware.
- **NFR-002 (Compatibility):** Fully certified for Android 8.0 (API level 26) through Android 14+.
- **NFR-003 (Reliability):** Zero-data-loss media uploading with automatic retry and Cloudinary secure CDN storage.

23. ACCEPTANCE CRITERIA

AC-ORD-001 (Successful Delivery)

- **Given:** The partner is within 600m of the destination coordinates,
- **And:** The partner has captured a matching face photo, 1-5 item photos, and entered the valid customer OTP,
- **When:** The partner taps 'Submit Delivery',
- **Then:** The system marks order as delivered, generates PDF invoice, syncs with Cloudinary & MongoDB, and opens the Native PDF Viewer.

24. EDGE CASES

- **Empty Input:** Submitting forms without completing required fields triggers localized validation messages.
- **Network Loss:** Dropped connectivity during upload prompts a retry dialog without clearing entered data.
- **No Registered Photo:** If beneficiary lacks a photo, the system prompts partner to register/update photo before delivery.
- **Camera Hardware Failure:** Catches camera exceptions gracefully and presents diagnostic guidance.

25. REQUIREMENTS TRACEABILITY MATRIX

Requirement ID	Module	Acceptance Criteria ID	Test Scenario ID	Test Case ID	Defect ID	Status
FR-AUTH-001	AUTH	AC-AUTH-001	TS-01	TC-01	-	Confirmed
FR-PROF-001	PROF	AC-PROF-001	TS-02	TC-02	-	Confirmed
FR-ORD-001	ORD	AC-ORD-001	TS-03	TC-05	-	Confirmed
FR-ORD-002	ORD	AC-ORD-001	TS-04	TC-06	-	Confirmed
FR-ORD-005	ORD	AC-ORD-002	TS-05	TC-08	-	Confirmed

26. ASSUMPTIONS

- Beneficiary records and initial allocations can also be onboarded directly via the mobile app dialog or administrative database.
- Cellular network is available at the start and completion points of the delivery transit.

27. CONSTRAINTS

- Facial verification requires sufficient ambient light to detect face landmarks via camera.
- Accurate GPS geofencing requires direct line of sight to navigation satellites (outdoors).

28. TECHNICAL EDGE CASES & ARCHITECTURAL CLARIFICATIONS

- 1. Partner Profile Onboarding (Postal & Address Resilience):** The Indian Postal API lookup implements an 8-second timeout. If the request fails, times out, or returns no records, the system displays an inline warning ('Unable to fetch location automatically') and activates a manual unlock toggle (edit-pencil icon on State field). Tapping this icon switches City & State to editable fields so operatives are never blocked. Address validation enforces a 5-character minimum and permits standard structural delimiters (e.g. '#', '/', '-', ',') to support Indian postal formats. Email validation blocks all whitespace characters and accepts all standard 2+ character top-level domains.
- 2. Authentication & Rate-Limiting:** The mobile authentication gateway enforces a threshold of 5 consecutive failed OTP attempts on POST /auth/verify-otp before a phone number is subjected to a 15-minute temporary lockout. Whenever an API endpoint returns HTTP 401 (Unauthorized) or a JWT token expires, the client immediately throws an UnauthenticatedException, purges session credentials from flutter_secure_storage, resets AuthProvider status to unauthenticated, and navigates to the Login screen.
- 3. Beneficiary Validation & Resource Authorization:** On-demand farmer creation requires Farmer Name (2-50 chars, alphabetic & spaces), 10-digit mobile number, Village, District, 6-digit PIN code, GPS coordinates, assigned items, and reference photo. If an operative tampers with an API payload to request a beneficiary assigned to another partner, the backend enforces scoped tenancy (delivery_partner_id filter) and returns HTTP 404 Not Found (Existence Masking) rather than 403, preventing resource enumeration.
- 4. Multi-Stage Handover Protocol (Biometrics, Video & Geofence):** If an SMS OTP fails to reach a beneficiary in remote terrain, the pre-registered OTP in the beneficiary record can be verified via supervisory clearance. For Gate 2 facial verification, Google ML Kit extracts landmark geometry and MobileFaceNet computes Euclidean distance across L2-normalized embeddings, requiring a strict distance < 0.90 to pass. If multiple faces appear in the camera frame, the algorithm isolates the largest face by bounding box area. Gate 4 video is restricted to a 30-second live MP4 capture streamed directly to Cloudinary. For Gate 5 geofencing, LocationAccuracy.high is requested; the 600m perimeter buffer is engineered specifically to absorb rural GPS drift under dense canopy.
- 5. Invoicing & PDF Synchronization:** Invoice PDFs are compiled locally on device storage via InvoiceService before network transmission to Cloudinary. If a network interruption occurs before receiving the Cloudinary CDN receipt URL, the app retains the local file path and delivery log data, allowing the operative to re-render, view, or share the invoice offline or retry upload without data loss.
- 6. Network Timeout & Retry Mechanics:** Streaming multipart uploads operate with a 120-second timeout. If an upload times out mid-transit due to cellular drops, the Order Completion form retains all in-memory file references (captured photo thumbnails and video path). The field operative does NOT need to re-capture any proofs; tapping 'Submit Delivery' immediately re-initiates the multipart payload.

29. TBD REQUIREMENTS

- Offline batch synchronization queue for zero-connectivity journeys.
- Push notification dispatch for newly scheduled village allocations.

30. REQUIREMENT QUALITY REPORT

Category	Count
Total Functional Requirements	16
Confirmed Requirements	14
Derived Requirements	2
Business Rules	4
Non-Functional Requirements	3
Quality Review Status	100% Passed (Version 1.2)